SKILLSYNC

Peer-to-Peer Skill Exchange Platform

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Short Version | Version 1.0 | October 2026

Prepared for: College Project Documentation

Frontend: Flutter | Backend and Database: To be finalized

1. Introduction

SkillSync is a peer-to-peer skill exchange platform that connects people who want to learn new skills with people who can teach them. Users can create profiles, list skills, discover suitable peers, send exchange requests, and share feedback.

1.1 Purpose

This SRS describes the main features, interfaces, and quality requirements of SkillSync. It helps developers, testers, and project stakeholders understand what the application should do.

1.2 Objectives

• Make peer-to-peer learning easier and more accessible.

• Help users find people with matching skills and interests.

• Support skill exchange requests and session planning.

• Encourage a safe and respectful learning community.

2. Overall Description

2.1 Product Perspective

SkillSync is planned as a mobile application. Flutter is the proposed frontend technology. A backend service and database will store user profiles, skills, requests, and feedback.

2.2 User Classes

• Guest: can register or log in.

• Registered User: can manage a profile, list skills, search peers, and exchange requests.

• Administrator: can review reports and manage inappropriate content.

2.3 Main Features

• Registration and login

• Profile and skill management

• Search and discover peers

• Send, accept, or reject exchange requests

• Arrange learning sessions

• Ratings, reviews, and notifications

• Report users or inappropriate content

3. Functional Requirements

ID	Requirement
FR-01	The system shall allow users to register, log in, and log out.
FR-02	Users shall be able to create and update their profiles.
FR-03	Users shall add skills they can teach and skills they want to learn.
FR-04	Users shall search and filter peers by skill or category.
FR-05	Users shall send, accept, reject, or cancel exchange requests.
FR-06	Users shall record session details and track exchange status.
FR-07	Users may submit ratings and reviews after an exchange.
FR-08	Administrators shall review reports and moderate content.

4. Non-Functional Requirements

• **Usability:** The interface should be simple and easy to understand.

• **Performance:** Common screens and searches should respond quickly under normal network conditions.

• **Security:** User accounts and private information must be protected by authentication and access control.

• **Reliability:** The system should save data correctly and show clear error messages.

• **Compatibility:** The app should work on the selected target mobile devices.

• **Maintainability:** The code should be organized into clear modules.

5. External Interface Requirements

• **User Interface:** Welcome, Login, Register, Home, Profile, Skills, Search, Requests, Session, Feedback, and Settings screens.

• **Software Interface:** The Flutter app will communicate with a backend API and database.

• **Hardware Interface:** A compatible smartphone or tablet is required. Photo access may be used for profile images.

• **Communication:** HTTPS should be used for secure network communication in deployment.

6. Data Requirements

The database is expected to store the following main entities:

• User: account and profile information.

• Skill: skill name, category, and description.

• UserSkill: skills offered and skills wanted by each user.

• ExchangeRequest: sender, recipient, request details, and status.

• Session: agreed date, time, mode, and completion status.

• Feedback: rating and review after a completed exchange.

7. Basic Use Cases

• Create Account: a guest registers for SkillSync.

• Manage Profile: a registered user updates personal details and skills.

• Find a Peer: a user searches for a person with a required skill.

• Request Exchange: a user sends a request and the other user responds.

• Complete Exchange: users arrange a session and mark it complete.

• Submit Feedback: participants share feedback after the session.

8. Assumptions and Constraints

• The frontend is planned using Flutter.

• A backend and database are required for shared, persistent data.

• Internet access is needed for online features.

• The first version focuses on peer skill exchange; paid courses and payment processing are outside the initial scope.

9. Acceptance Criteria

• Users can register and log in successfully.

• Users can add skills offered and skills wanted.

• Search results show relevant peers.

• Exchange requests can be sent and their status updated.

• The app saves user and request information.

• Unauthorized users cannot access administrator functions.

10. Conclusion

SkillSync aims to make learning more collaborative by helping people exchange skills directly. This SRS provides a short summary of the requirements for developing and testing the initial version of the application.